



UNIVERSITY OF
CENTRAL FLORIDA

CAP 4611 - ALG. FOR MACHINE LEARNING

Section: C001

*College of Engineering and
Computer Science*

Department of Computer Science

Course Information

Term: Summer 2026

Class Meeting Days: TR

Class Meeting Time: 12:00PM - 01:50PM

Class Meeting Location: CB2 O101

Modality: P

Credit Hours: 3.00

Instructor Information



Tanvir Ahmed

Title: Associate Lecturer

Office Location: HEC 219

Office Hours:

Monday 12:30 pm - 2:30 pm (Zoom)

Wed 12:30 pm - 2:30 pm (Zoom)

(Check the first module on Webcourses for my Zoom link)

Phone: 4078235043

Email: Tanvir.Ahmed@ucf.edu

Communication Expectations

Important Email and Communication Policy

I have three large classes this semester, so please follow the communication and email policy below to help ensure effective communication. As noted below, if you do not receive a response from me by email, **please visit me during office hours.**

Important Email Policy (**Please, do not send Webcourses messages**)

Please read and follow the communication guidelines below when contacting me or the TAs.

1. Use your UCF email account for all communication.
Do NOT send messages through Webcourses.
2. If you are following up on a previous email, always use the **same email thread**.
3. In the subject line of every email, include:
 - the course name,
 - section number, and
 - a brief description of the topic.Since I teach multiple courses and sections, this helps us respond more efficiently.
4. Emails should be reserved for **administrative inquiries only** and should remain brief, professional, and friendly.
5. Questions about course content should be discussed during office hours, where we can have a more effective and interactive discussion. We may not help with debugging your code in email.
6. My direct email address should only be used for matters that are:
 - not relevant to the TAs, or
 - intended to remain private from the TAs.Examples include illness, family emergencies, SAS accommodations, or private discussions regarding course performance.
7. I generally respond to emails within **48 hours**(excluding weekends). If you do not receive a response within that timeframe:
 - reply to the same email thread as a follow-up, or
 - Visit me during office hours or the Zoom office hour meeting.
8. Due to the large volume of emails in this course, occasional missed emails can happen. In urgent situations, please follow up using the same email thread.

Contacting the TAs

In general, do not request assignment-related help through email. We offer many office hours specifically for assignment assistance.

For grading inquiries, please follow the grading inquiry process described later in this syllabus.

For general questions, or if you are unsure which TA to contact:

- send the email to **all TAs assigned to your section**,
- place the TA email addresses in the "To" field, and
- place the instructor email address in the "CC" field.

The TA responsible for your question will respond to you.

As always, remember to include the **course name, section number, and topic** in the email subject line.

Course Description

CAP 4611 ENGR-COMP SCI 3(3,0) Algorithms for Machine Learning: PR: Earn a minimum grade of C in each of the following: COP3502C - Computer Science I (3) STA2023 - Statistical Methods I (3) An overview of the most commonly used algorithms for supervised, unsupervised, and reinforcement learning.

Introduction to experimental design, evaluation metrics, and applications of machine learning.

This course introduces the fundamental algorithms and techniques used in machine learning, with an emphasis on experimental setup, how models learn from data, and optimization methods. Topics include exploratory data analysis, model training and evaluation, and the practical implementation of machine learning algorithms using Python and scikit-learn. Students will study supervised and unsupervised learning methods such as linear and logistic regression, decision trees, Naive Bayes, support vector machines, k-nearest neighbors, clustering, ensemble methods, association rule learning, and selected natural language processing techniques. The course also provides an introductory treatment of neural networks and convolutional neural networks, with hands-on exposure using modern frameworks such as Keras.

Student Learning Outcomes

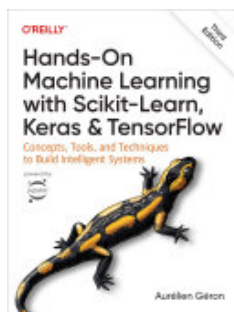
Upon successful completion of this course, students will be able to:

1. **Explain** the fundamental principles of machine learning, including supervised and unsupervised learning paradigms.
2. **Apply** optimization techniques to train machine learning models and analyze their convergence behavior.
3. **Implement** core machine learning algorithms (e.g., regression, decision trees, SVM, k-NN, clustering, ensemble methods) using Python and scikit-

learn.

4. **Perform** exploratory data analysis and data preprocessing to prepare real-world datasets for machine learning tasks.
5. **Design and evaluate** experimental setups, including proper data splitting, cross-validation, and performance metrics.
6. **Analyze and compare** machine learning models using appropriate evaluation measures and justify algorithm selection.
7. **Apply** machine learning techniques to selected application domains such as text data (basic NLP) and association rule learning.
8. **Demonstrate** foundational understanding of neural networks and convolutional neural networks using high-level frameworks such as Keras.
9. **Communicate** machine learning results effectively through written reports and visualizations.

Required Course Materials and Resources



Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow

ISBN: 9781098122461

Authors: Aurélien Géron

Publisher: "O'Reilly Media, Inc."

Publication Date: 2022-10-04

Notes and additional resources will be available on webcourses

Other important tools and materials

Data camp

To help everyone get up to speed with the coding side of machine learning, I have secured access to DataCamp for the semester. Everyone should be able to join using their **ucf.edu** email address. To ensure access, do not create an account prior to clicking the join link.

The link is available on webcourses module

*DataCamp lessons supplement the textbook readings and **will be a required component of this course in the first 3 weeks.** [Expect to spend 15+ hours per week for the first two weeks on Datacamp Python and data analysis library courses]. I have also linked various data camp courses that you can do for more learning. However, those extra courses are optional*

Additional Resources

In addition to the DataCamp material, the following content on python, pandas, scikit-learn, and tensorflow will be critical to your success in this course:

- <https://www.kaggle.com/learn/python>
- <https://www.kaggle.com/learn/pandas>
- <https://docs.python.org/3/tutorial/index.html>
- <https://docs.python.org/3/reference/index.html>
- <https://docs.python.org/3/library/index.html>
- https://pandas.pydata.org/pandas-docs/stable/getting_started/index.html
- https://pandas.pydata.org/pandas-docs/stable/user_guide/10min.html
- https://pandas.pydata.org/pandas-docs/stable/user_guide/index.html
- https://scikit-learn.org/stable/getting_started.html
- <https://scikit-learn.org/stable/tutorial/index.html>
- https://scikit-learn.org/stable/user_guide.html
- <https://www.tensorflow.org/tutorials>

Related Courses

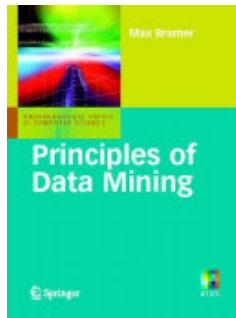
- [Machine Learning and Data Mining \(UBC\)](#)
- [Introduction to Machine Learning \(Alberta\)](#)
- [Practical Machine Learning \(Berkeley\)](#)
- [Machine Learning \(MIT\)](#)
- [Machine Learning \(CMU\)](#)
- [Course in Machine Learning \(Maryland\)](#)
- [Data Mining \(CMU\)](#)

For the necessary mathematical background, Khan Academy and MIT have a number of useful courses:

- <https://www.khanacademy.org/math/differential-calculus>
- <https://www.khanacademy.org/math/integral-calculus>

- <https://www.khanacademy.org/math/calculus-1>
- <https://www.khanacademy.org/math/calculus-2>
- <https://www.khanacademy.org/math/multivariable-calculus>
- <https://www.khanacademy.org/math/statistics-probability>
- <https://www.khanacademy.org/math/linear-algebra>
- <https://ocw.mit.edu/courses/mathematics/18-01sc-single-variable-calculus-fall-2010/>
- <https://ocw.mit.edu/courses/mathematics/18-02sc-multivariable-calculus-fall-2010/>
- <https://ocw.mit.edu/courses/mathematics/18-06sc-linear-algebra-fall-2011/>
- <https://ocw.mit.edu/courses/mathematics/18-065-matrix-methods-in-data-analysis-signal-processing-and-machine-learning-spring-2018/>
- <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-041sc-probabilistic-systems-analysis-and-applied-probability-fall-2013/>

Recommended Course Materials



Title: Principles of Data Mining

ISBN: 9781846287657

Authors: Max Bramer

Publisher: Springer

Publication Date: 2007-03-28

Additional Information: Some notes will also refer this book

Course Assessment and Grading Procedure

The **tentative point** distribution will be as follows. Depending on the course flow and the situation, the course instructor may change the weights for future items. Any changes will be announced on webcourses or in the class:

Grading items	Weight
Syllabus quiz	0.5%
Programming and ML Assignments Jupyter Notebooks	55%
In Person Quizzes (best 3 out of 4) - must bring a laptop	9% [3 for each]
Practice topic-wise exercise quizzes online	4%

Datacamp course completion certificates for the assigned courses for learning Python related to data science operations	4.5%
In Class Midterm exam (must bring a laptop)	12%
In Class Final Exam (must bring a laptop)	15%
Total	100%

Final Exam Policy: You must attend the final exam and earn at least 35% on it to be eligible for a grade of C or higher in the course. This means that even if your overall course grade is above 66%, you will not receive a C if you score less than 35% on the final exam.

Grading Scale

Letter Grade	Percentage
A	100% - 90%
A-	< 90% - 86%
B+	< 86% - 82%
B	< 82% - 78%
B-	< 78% - 74%
C+	< 74% - 70%
C	< 70% - 66%
C-	< 66% - 62%
D+	< 62% - 58%
D	< 58% - 54%
D-	< 54% - 50%
F	< 50% - 0%

Policies for Course Grade

Programming/ML Assignments and Google Colab (Read the policy!)

Programming and machine learning assignments will be posted on Webcourses and must be completed **individually**. Students are expected to submit **original**

work and must comply with the AI usage policy described below. All assignments must be implemented in **Python** and submitted as a **Google Colab link**. Once an assignment is submitted, **no modifications to the Google Colab notebook are permitted**.

If you have questions regarding an assignment, you are encouraged to seek help from the instructor or a teaching assistant during office hours, post questions on the discussion board, or ask general questions in the course Discord server (TAs may participate occasionally). **Under no circumstances may you share your code**, in part or in full, with other students in this course. If you wish to assist classmates, you may share **learning resources or conceptual guidance**, but not code or solution details.

Late assignments will not be accepted via email.

In-person Exams and Quizzes (Read the policy!)

- **Exams will test your understanding of the various concepts discussed in the class, such as understanding of various ML algorithms and their use cases, various parameters, data analytics concepts, general ML concepts, cost functions, math behind them, etc. The quizzes and exams will NOT test your coding an implementation of ML model.**
- **All 4 quizzes and exams are in-person (It is an in-person class!)**
- **You must bring a laptop to take those exams/quizzes in person in the classroom. Attempting to take the tests outside of the classroom will be considered as cheating and will result in F grade in the class and may be reported to the Office of Student Conduct**
- **Access code will be provided in the class and you must not share the access code with others**
- **All exams are closed-books, closed-notes, closed-neighbors, close AI, etc. You are not allowed to open any tab/tabs on your computer other than webcouses quiz you are taking for the tests.**
- ***Manage your time efficiently during the exam!!!***

Makeup Work Policy

Make-up exams and submissions will be very limited. Make-up exams will be given in the event of university-excused absences. Excused absences, as defined (or at least suggested) by the Academic Regulations and Procedures governing the university, may include: “serious illness, serious family emergencies, special curricular requirements (e.g., judging trips, field trips, and professional conferences), military

obligations, severe weather conditions, and religious holidays.” Furthermore, “participation in official University-sponsored activities, such as music performances, athletic competition, or debate” constitutes an excused absence.

Note that living far from the campus, personal vacation, doing job during class time, family gathering events, job interviews, etc. are not excused reasons to request makeup. You should not schedule anything during the class time as you have registered for this class keeping the class schedule in mind.

In the case of planned absences, make-up exams must be arranged with the instructor prior to the absence in question. Of course, some excused absences cannot be foreseen (e.g., illness or serious family emergencies). In these cases, you must contact the instructor as soon as possible (within 24 hours) to schedule a makeup exam. **If you do not contact the instructor in a timely manner (typically within 24 hours of missing the exam), you may be denied a make-up exam.**

Assignment deadlines may be extended in these situations as well, or, in the case of planned absences, you may be required to turn the assignment in early or on time, depending on the nature of the assignment and the circumstances of the absence; this decision is at the discretion of the course instructor. Your best course of action is to inform me as early as possible if you know you’re going to be out of town or missing classes for an excused reason!

Make-up exams and deadline extensions may be granted in other severe situations as well, at the discretion of the instructor. TAs are unable to give extensions. All such requests must go through the course instructor.

Note that if I made a solution available to all the students for any assignment, then the excused extended deadline will not work for the same assignment. The instructor will assign a different problem in that case.

Attendance and class participation

There is no grade for attendance. However, attending and actively participating in the class is critical to grasp the concepts and prepare for the exams. Although there is no attendance grade, I will randomly take attendance in different classes for my recording purpose and it will be taken within the first 2-3 minutes of the class. Moreover, in case of very low attendance regularly, I might announce in the class about new attendance policy and introduce the attendance grade. It might not be announced on the webcourses. I also encourage you to bring a laptop to the class. For attendance tracking, we will use UCF Here web app: <https://here.cdl.ucf.edu/>

Make-up Assignments for Authorized University Event or Co-curricular Activities

Students who represent the university at an authorized event or activity (for example, student athletes) and who are unable to meet a course deadline due to a conflict with that event must provide the instructor with documentation in advance to arrange a make-up. No penalty will be applied if the student gives advance notice and communicates with the instructor following UCF policy. In the case of an authorized university activity, it is your responsibility to show the instructor a signed copy of the Program Verification Form for which you will be absent, prior to the class in which the absence occurs or due date you need extended. For more information, see [UCF Policy 4-401](#).

Make-up Assignments for Religious Observances

A student who desires to observe a religious holy day of his or her religious faith must notify their instructor as soon as practicable prior to the observance. Students who are absent because of religious observances will be permitted a reasonable amount of time to complete any missed work. For more information, see [UCF Regulation 5-020](#).

Artificial Intelligence (AI) Use Policy

Use of Generative Artificial Intelligence (GenAI) is prohibited. Use of GenAI tools via website, app, or any other access, is not permitted in this class. All components of assignments in this course must be independently and originally completed by the student. Representing work created by GenAI as your own work will be treated as plagiarism.

Guidelines to be successful in this course

Here are some guidelines to be successful in this course:

- Review prerequisite knowledge in data structures, algorithms, basic probability, and linear algebra as needed.
- Be proactive in learning Python; consistent practice outside class is essential.

- Complete all programming assignments early to allow time for debugging and experimentation.
- **Focus on understanding *how* and *why* machine learning algorithms work, not just how to run them.**
- Actively participate in lectures and discussions, and ask questions when concepts are unclear.
- Review the notes thoroughly after the lecture
- Practice exploratory data analysis and visualization to **better understand datasets before modeling.**
- **Carefully analyze experimental results and use appropriate evaluation metrics.**
- Use office hours, TA sessions, and discussion forums regularly, do not wait until deadlines.
- Follow academic integrity policies and submit original work.
- Manage time effectively, as machine learning assignments often require iterative refinement.

Grading Inquiries and review your exams

Every now and then, someone on my team will make a minor error while grading an assignment or an exam. If you want to revieve your quiz or exam or you think your work is mis-graded, you must bring the matter to me and the TA for resolution within 5 days of the assignment or exam having been graded (Except final exam and last assignment. They have to be inquired within 6 hours after receiving your grade). In general, you need to pick a face to face office hours of the TA who graded this and inform the TA about it 24 hours before the office hour so that the TA can bring the paper with them. If you want to review the paper with me, then let them know as well so that they can send a scanned copy to me and we can review the paper together during my office hour. In case inquiries for non-exam grading items, you should be able to see which TA has graded it. **You should write that TA an email and put me in the cc** of the email with the inquiry. If you are unable to determine which TA has graded it, send all the TA of your section an email and put me in the cc as discussed in the email policy. If you do not hear back from us for a grading inquiry within 48 hours, please follow-up in the same email thread. *[Again, I would like to remind you that, do not use webcourses message while writing to us. Use only your UCF email].*

Disability Access & Accommodations

The University of Central Florida is committed to providing equal access to all students with disabilities (ADHD, learning disabilities, Autism, chronic medical

conditions, physical disabilities, etc.). To receive consideration for reasonable disability-related course accommodations, disabled students must contact Student Accessibility Services (SAS) and complete the steps required for SAS to review accommodation requests. More information can be found on the UCF [Student Accessibility Services](#) website under the Start Here tab or by contacting SAS directly (Ferrell Commons 185; sas@ucf.edu; Phone - 407-823-2371).

Approved accommodations are shared with course instructors via the SAS Course Accessibility Letter. Implementing certain accommodations may require discussion and specific considerations of the course design, course learning objectives, and the individual academic and course challenges experienced by the student. While students with disabilities or chronic health needs are also encouraged to discuss any course concerns with professors in addition to contacting SAS, professors are not required to facilitate disability-related adjustments to the course unless the professor has received a Course Accessibility Letter from SAS that outlines approved accommodations.

Academic Integrity

Students should familiarize themselves with UCF's Code of Conduct at [Student Conduct and Integrity Office](#). According to Section 1, "Academic Misconduct," students are prohibited from engaging in:

- a. Academic misconduct is defined as any submitted work or behavior that obstructs the instructor of record's ability to accurately assess the student's understanding or completion of the course materials or degree requirements (e.g., assignment, quiz, and/or exam). Examples of academic misconduct include but are not limited to: plagiarism, unauthorized assistance to complete an academic exercise; unauthorized communication with others during an examination, course assignment, or project; falsifying or misrepresenting academic work; providing misleading information to create a personal advantage to complete course/degree requirements; or multiple submission(s) of academic work without permission of the instructor of record.
- b. Any student who knowingly helps another violate academic behavior standards is also in violation of the standards.
- c. Commercial Use of Academic Material. Selling of course material to another person and/or uploading course material to a third-party vendor without authorization or without the express written permission of the

University and the instructor of record. Course materials include but are not limited to class notes, the instructor of record's slide deck, tests, quizzes, labs, instruction sheets, homework, study guides, and handouts.

- d. Soliciting assistance with academic coursework and/or degree requirements. The solicitation of assistance with an assignment, lab, quiz, test, paper, etc., without authorization of the instructor of record or designee is prohibited. This includes but is not limited to asking for answers to a quiz, trading answers, or offering to pay another to complete an assignment. It is considered Academic Misconduct to solicit assistance with academic coursework and/or degree requirements, even if the solicitation did not yield actual assistance (for example, if there was no response to the solicitation).

Responses to Academic Dishonesty, Plagiarism, or Cheating

Students should also familiarize themselves with the procedures for academic misconduct in UCF's student handbook, [The Golden Rule](#). UCF faculty members have a responsibility for students' education and the value of a UCF degree, and so seek to prevent unethical behavior and respond to academic misconduct when necessary. Penalties for violating rules, policies, and instructions within this course can range from a zero on the exercise to an "F" letter grade in the course. In addition, an Academic Misconduct report could be filed with the Office of Student Conduct and Academic Integrity, which could lead to disciplinary warning, disciplinary probation, or deferred suspension or separation from the University through suspension, dismissal, or expulsion with the addition of a "Z" designation on one's transcript.

Being found in violation of academic conduct standards could result in a student having to disclose such behavior on a graduate school application, being removed from a leadership position within a student organization, the recipient of scholarships, participation in University activities such as study abroad, internships, etc.

Title IX

Title IX prohibits sex discrimination, including sexual misconduct, sexual violence, sexual harassment, and retaliation. If you or someone you know has been harassed or assaulted, you can find resources available to support the

victim, including confidential resources and information concerning reporting options at [Let's Be Clear](#) and [UCF Cares](#).

For more information on access and community engagement, Title IX, accessibility, or UCF's complaint processes contact:

- Title IX – ONAC – [Office of Nondiscrimination & Accommodations Compliance](#) & askanadvocate@ucf.edu
- Disability Accommodation – Student Accessibility Services – [Student Accessibility Services](#) & sas@ucf.edu
- [Access and Community Engagement](#) (including the Ginsberg Center for Inclusion and Community Engagement, Military and Veteran Student Success, and HSI Initiatives)
- UCF Compliance and Ethics Office – [Compliance, Ethics, and Risk Office](#) & complianceandethics@ucf.edu
- The [Ombuds Office](#) is a safe place to discuss concerns.

Reporting an Incident or Issue

If you believe you have experienced discrimination by any faculty or staff member, contact the Office of Nondiscrimination & Accommodations Compliance via the [ONAC website](#) or at 407-823-1336. You can also choose to report using the UCF Integrity Line either anonymously or as yourself at 1-855-877-6049 or by using the [online form](#). UCF cares about you and takes every report seriously. For more information see the [Reporting an Incident or Issue Webpage](#).

Deployed Active-Duty Military Students

Students who are deployed active duty military and/or National Guard personnel and require accommodation should contact their instructors as soon as possible after the semester begins and/or after they receive notification of deployment to make related arrangements.

Campus Safety

At UCF Public Safety and Police, safety is the top priority. Emergencies on campus are rare, but if one should arise, it's important to be familiar with some basic safety and security concepts.

- In an emergency, always dial 911.
- Every UCF classroom has an **Emergency Procedure Guide** posted on a wall near the door, which will show you how to respond to a variety of

situations. This guide can also be found online [here](#).

- In the event of an active threat, remember **AVOID, DENY, DEFEND**. Choose the best course of action and act immediately. Watch the video [here](#) to learn more.
 - **AVOID**. Pay attention to your surroundings and have an exit plan. Get as much distance and as many barriers between you and the threat as quickly as possible.
 - **DENY**. When avoiding is difficult or impossible, deny the threat access to you and your space. Lockdown by creating barriers, turning the lights off and remaining quiet and out of sight. Make sure your cell phone is silenced, but do not turn it off.
 - **DEFEND**. When you are unable to put distance between yourself and the threat, be prepared to protect yourself. Commit to your actions, be aggressive and do not fight fairly. Do whatever it takes to survive.
- For emergencies on campus, UCF will utilize the [UCF Alert](#) system. All UCF students, faculty and staff are automatically enrolled to receive these email and text alerts, however, it's a good idea to frequently ensure your [contact information is up to date](#).

Financial Aid Accountability

All instructors/faculty are required to document students' academic activity at the beginning of each course. In order to document that you began this course, please complete this activity by the end of the first week of classes or as soon as possible after adding the course. Failure to do so may result in a delay in the disbursement of your financial aid.

Class Schedule

Tentative Schedule:

The following table presents the **tentative** schedule for lectures, quizzes, assignments, and exams. Depending on circumstances, some lectures may be conducted earlier or carried over to subsequent sessions. Any changes to the assignment or quiz schedule will be announced both in class and on Webcourses. Students are advised to check Webcourses regularly for exercise and lab submission deadlines, as these may vary by time of day. Additional details are provided in the exercise submission policy.

Week	DATE	Lecture Topic	Suggested Optional Reading/video	Related Assignments
1	May 11 – May 17	No class on Tuesday (May 12) First lecture Thursday, May 14) Course structure and syllabus discussion; Introduction: Introduction of various concepts of ML as an overview, such as definition and examples of ML and use cases, ML process, types of ML problems, supervised, unsupervised, reinforcement, various ways of learning, challenges of ML, overfitting/underfitting, regularization, testing and validation	Chapter 1 [HOML] Chapter 2 [HOML] Notes: Check Module	Academic Engagement Quiz Week 1 Datacamp Datacamp Data Science Track: <i>Required:</i> • Introduction to Python • Intermediate Python • Python for Data Science Toolbox (Part 1) • Data Manipulation with Pandas • Introduction to Data Visualization with Matplotlib <i>Optional:</i> ◦ Python for Data Science Toolbox (Part 2) ◦ Joining Data with Pandas <i>(Going through the Iterator topic is strongly recommended)</i>

2	May 18–May 24	Continue the remaining parts from the introduction; <u>Exploratory Data Analysis:</u> Necessity of data exploration, ML workflow, data types, feature extraction, data quality issues, plotting and visually explore relationships between variables, sampling techniques, Coding for data exploration	Chapter 3 [HOML] Chapter 4 [HOML] Notes: Check Module Datacamp Data Science Track: • Statistical Thinking Python (Part 1) • Statistical Thinking Python (Part 2)	Week 2 data camp Datacamp Data Science Track: <i>Required:</i> • Introduction to Data Visualization with Seaborn • Intermediate Data Visualization with Seaborn • Data Cleaning in Python <i>Optional:</i> • Introduction to Importing Data in Python • Intermediate Importing Data in Python • Working with Dates and Times in Python • Writing Functions in Python
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3	May 25–May 31	Continue data exploration. <u>Classification and evaluation:</u> definitions and various types of classification problems, strategies for model performance evaluations, and various matrices, cross-validation, dealing with time series data,	Chapter 5 [HOML] Chapter 6 [HOML] Notes: Check Module Datacamp Machine Learning Scientist Track: • Supervised Learning with scikit-learn • Preprocessing for Machine Learning in Python • Feature Engineering for Machine Learning in Python • • Model Validation in Python	Week 3 Datacamp ◦ Required: • Exploratory Data Analysis in Python
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4	June 01 – June 07	<u>In class quiz 1 on Tuesday.</u> Continue Classification and Evaluatoin; Linear Regression; Gradient descent, stochastic gradient descent and related topics,	• Chapter 7 [HOML] • Chapter 8 [HOML] • Datacamp Machine Learning Scientist Track: ◦ Linear Classifiers in Python ◦ Machine Learning for Time Series Data in Python ◦ Hyperparameter Tuning in Python	<u>Assignment 1 due: Exploratory Data Analysis using Python (alert: lengthy assignment).</u> <u>In class quiz 1 on Tuesday;</u>
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5	June 08 – June 14	Continue regression models; polynomial regression and regularized models Logistic regression;	- Chapter 9 [HOML] - Chapter 10 [HOML] - Datacamp Machine Learning Scientist Track: Machine Learning with Tree-Based Models in Python Winning a Kaggle Competition in Python	
6	June 15 – June 21	Continue logistic regression; softmax regression; KNN Classifier;	- Chapter 11 [HOML] - Chapter 12 [HOML] - Datacamp Machine Learning Scientist Track: Extreme Gradient Boosting with XGBoost	

7	June 22 – June 28	<u>In person quiz 2 on Tuesday;</u> Naive Bayes classifier; <u>In-class midterm exam on Thursday,</u>	- Chapter 13 [HOML] Datacamp Machine Learning Scientist Track: Dimensionality Reduction in Python	<u>In person quiz 2 on Tuesday;</u>
8	June 29 – Jul 05	Support Vector Machines; Decision Tree; Ensemble Learning, Random Forests, and Boosting	- Chapter 14 [HOML] Datacamp Machine Learning Scientist Track: Unsupervised Learning in Python Cluster Analysis in Python	

9	Jul 06 - Jul 12	Dimensionality Reduction and PCA Unsupervised Learning: Clustering	- Chapter 15 [HOML] http://d2l.ai/chapter_linear_networks/index.html http://d2l.ai/chapter_multilayer_perceptrons/index.html https://www.deeplearningbook.org/contents/mlp.html Datacamp Machine Learning Scientist Track: - Introduction to TensorFlow in Python	
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10	Jul 13 - Jul 19	In person Quiz 3 Tuesday Natural language processing (NLP); Introduction to neural network	Chapter 16 [HOML] http://d2l.ai/chapter_deep-learning-computation/index.html https://www.deeplearningbook.org/contents/regularization.html https://www.deeplearningbook.org/contents/optimization.html Datacamp Machine Learning Scientist Track: • Introduction to Deep Learning in Python • Introduction to Deep Learning with Keras	Assignment 4 due Sunday: Clustering (Medium length assignment)
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11	Jul 06 - Jul 12	Convolutional Neural Networks and Keras; Association Rule Mining Quiz 4 Thursday (may move to next Tuesday as well)	- Chapter 17 [HOML] http://d2l.ai/chapter_convolutional-neural-networks/index.html http://d2l.ai/chapter_convolutional-modern/index.html https://www.deeplearningbook.org/contents/convnets.html Datacamp Machine Learning Scientist Track: Image Processing in Python Image Processing with Keras in Python	Assignment 5: NLP and deep learning (Shorter assignment) Quiz 4 thursday (may move to next Tuesday as well)
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12	July 27 – Aug 02	Transformer and attention mechanism(if we get time) Final Exam- During the last lecture on Thursday, Jul 30, 2026	• http://d2l.ai/chapter_recurrent-neural-networks/index.html • http://d2l.ai/chapter_recurrent-modern/index.html • https://www.deeplearningbook.org/contents/rnn.html Datacamp Machine Learning Scientist Track: ◦ Introduction to Natural Language Processing in Python ◦ Feature Engineering for NLP in Python ◦ Datacamp Machine Learning Scientist Track: ▪ Advanced Deep Learning	Assignment 6 due on Sunday: CNN (Shorter assignment) Final exam Thursday
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			g_with Keras	
	<u>Grades due by Wednesday, August 5, 2026 12:00 PM (Noon).</u>			

* HOML = Hands-on Machine Learning

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